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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING / SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2021-2022

Programme: B. Tech / MBA Tech (All Streams) Year: I Semester: II

Subject: Data Structures and Algorithms

Date: 04 July 2022

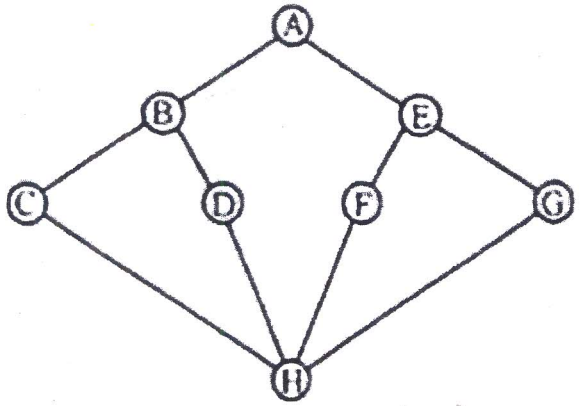
Marks: 100
Time: 11.00 am to 2.00 pm
Durations: 3 (Hrs)
No. of Pages: 02

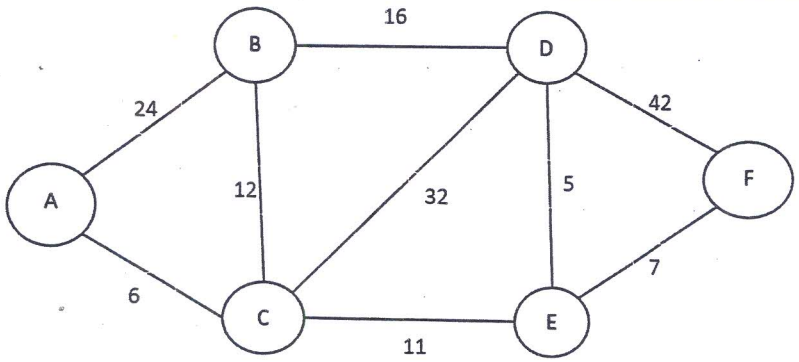
Re-Examination(2021-22)

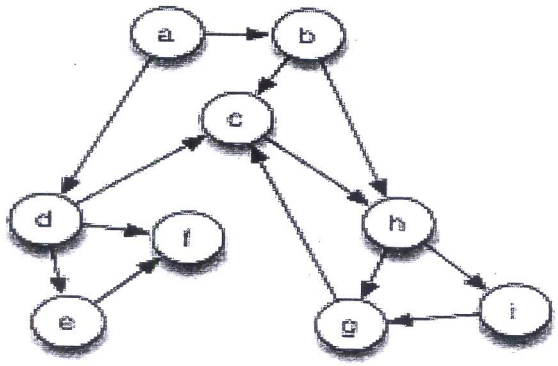
Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary**

Q1		Answer briefly:	
CO- 1 SO- 1 BL- 2	a.	Identify the most suitable data structure for the given scenario. Provide your justification for the same. I. An "Undo" mechanism in text editor II. Nested function calling in programming language III. To store directories and files in a computer. IV. Representation of friendships on social networking sites V. Keeping a track of emails based on importance.	[5]
CO- 2 SO- 1 BL- 2	b.	Write a function to delete a last node from a singly linked list.	[5]
CO- 3 SO- 1 BL- 3	c.	Perform depth first search traversal for the given graph. Show all the steps with suitable data structure. [Assume node **A** as a root node]	[5]

			
CO- 4 SO- 1 BL- 3	d.	Explain difference between linear search and binary search. For the given values in an array, demonstrate use of binary search to find an element 88 from an array. Index: 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 Values: 12 | 25 | 32 | 47 | 49 | 67 | 88 | 93	[5]
Q2			
CO-2 SO- 1 BL- 3	a.	The roll numbers of students participating in an event are to be stored in a singly linked list. The node will store the roll number of student. Write functions to perform below operations on the linked list. 1. Insert a roll number of a student at the particular position given by the user. 2. Delete a roll number of a last student in a linked list.	[10]
CO-3 SO- 1 BL-6	b.	Construct a binary search tree using the following data elements and answer the below questions. The elements are inserted in the order given. 1. Perform pre-order traversal of the constructed tree 2. Re-draw the tree after deletion of 57 3. Find out height of the tree 45, 39, 57, 11, 33, 79, 31, 9, 90, 55, 66, 80	[10]
Q3			
CO-3 SO- 1 BL- 2	a.	Calculate shortest path from node A to all other nodes in the following graph using Dijkstra's algorithm. **Show all the steps.**	[10]

			
CO-1,4 SO- 1 BL- 2	b.	Explain Big Oh notation and write an algorithm to sort an array using selection sort.	[10]
Q4			
CO-2 SO- 1 BL- 2	a.	Write an algorithm to perform dequeue operation on a circular queue. Identify and draw a circular Queue position in below scenarios. Assume size of the Queue as 6. 1. Front ==Rear 2. Front==0 and Rear==size-1 3. Front==Rear+1 4. Front<Rear 5. Front==-1 and Rear==-1	[10]
CO-4 SO - 1 BL- 2	b.	Write a program to search a key element from an array using binary search technique.	[10]
Q5			
CO-2 SO- 1 BL-3	a.	Consider the following arithmetic expression. Convert the following expression to postfix expression-using stack. **Show all steps.** A+ (B*C-(D/E/F)*G)*H Evaluate the converted expression-using stack for the given values. A=5, B=15, C=2, D=6, E=3, F=2, G=5, H=4	[10]
CO-3 SO- 1 BL- 2	b.	Write functions to perform below operations on binary search tree. 1. To search a node from a binary search tree 2. To perform post-order traversal	[10]
Q6			

CO-4 SO- 1 BL- 2	a.	A certain sorting technique is applied to the following data set in ascending order: 81, 72, 63, 45, 27, 36 After two iterations, the rearrangement of the data set is given as below: 63, 72, 81, 45, 27, 36 Identify the sorting algorithm that is applied and complete the remaining passes. Write an algorithm of identified sorting technique	[10]
CO-2,3 SO- 1 BL- 3	b.	Demonstrate use of stack data structure to find depth first search sequence for the given graph. Write a function to perform depth first search traversal. 	[10]